Sayan Ghosh

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Summary

Final-year B.Tech CS (Cybersecurity) student at VIT Bhopal with hands-on experience building production-grade distributed systems and backend services in Go. Comfortable working across the stack from low-level C++ data structures and custom TCP protocols to REST APIs, PostgreSQL, and Docker-based microservices. Interested in backend engineering, systems programming, and applied security.

Education

VIT Bhopal University — **Aug. 2023 – May 2027**
B.Tech, Computer Science (Cybersecurity) | CGPA: 7.88/10 — *Bhopal, MP, India*

CBSE Board (Class XII) — **April 2023**
Science Stream (PCM+CS)

Technical Skills

Primary: Go (Golang), C++17, Python, JavaScript (Node.js / ES6+), Java
Frontend: React.js, TanStack Query, Recharts, HTML5, CSS3, Tailwind CSS
Backend & APIs: Go Microservices, REST API Design, Socket.io, Node.js / Express
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis
DevOps & Tools: Docker Compose, Git, Linux
Cloud: AWS EC2, Lambda, S3

Projects

Distributed Rate Limiting System | *Go, C++17, Docker, POSIX Sockets* — **GitHub**

- Built a Go API gateway that enforces per-client rate limits by communicating with a C++ backend over a custom binary protocol on raw POSIX TCP sockets, avoiding HTTP overhead entirely.
- Implemented a Scalable Counting Bloom Filter in C++17 using the Kirsch-Mitzenmacher double-hashing optimization to track request fingerprints with configurable false-positive rates.
- Completed binary framing protocol, Bloom filter integration, and full Docker Compose orchestration across the Go gateway and C++ backend; custom binary protocol chosen over HTTP to eliminate serialization overhead on the hot path.

Home Lab Threat Intelligence & Detection Dashboard | *Go, Python, PostgreSQL, Docker, Scapy* — **GitHub**

- Engineered a real-time network security monitoring system using a decoupled microservices architecture: Python/Scapy for packet capture, a high-throughput Go REST ingestion API, and PostgreSQL for structured storage.
- Implemented an unsupervised ML anomaly detection pipeline (Isolation Forest) that automatically flags suspicious traffic patterns including potential C2 communication.
- Developed a SOC-style operational dashboard in Streamlit with a 5-second refresh loop querying live packet logs for instant threat visibility.
- Containerised the entire multi-service stack with Docker Compose; designed PostgreSQL schema with custom INET types for efficient IP-level querying.

Ad Event Ingestion Engine | *Go, PostgreSQL, Redis, React, Docker, k6* — **GitHub**

- Designed a high-throughput ad-event ingestion service in Go (chi, pgx, Redis) sustaining ∼5,000 events/sec (295K events in 60s, 100% success, 0 dropped, ∼1ms median latency) via a buffered-channel goroutine worker pool with batched pgx.CopyFrom inserts to PostgreSQL; load-tested with k6.
- Hardened after a production review: fixed an unbounded per-IP rate-limiter (OOM risk) with TTL eviction, eliminated a cache stampede by coalescing concurrent misses via singleflight (50 concurrent reads → 1 DB query), and added hourly rollup tables for efficient metric aggregation.
- Shipped 17 race-tested Go unit/integration tests, a Dockerized multi-stage scratch image, graceful shutdown, and a React + TanStack Query + Recharts dashboard with live backpressure visualization.

Certifications & Achievements

- Supervised Machine Learning – Coursera / DeepLearning.AI (Andrew Ng)
- Google IT Support Professional Certificate – Google/Coursera
- Oracle Cloud Infrastructure Certified Data Science Professional – Oracle
- Solved 200+ DSA problems across LeetCode and coding platforms